

CDS DS 110: Introduction to Data Science with Python

Instructor Names: Dr. Scott Ladenheim (A1)
Dr. Kevin Gold (B1)

Course Dates: 9/3-12/19

Times and Locations: A1 MWF 12:20-1:10 SHA 110 Course Credits: 4
B1 MWF 1:25-2:15 CGS 511

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Course Description:

DS 110 is the first in a two-course sequence (including DS 210) that builds students' skills in computing techniques central to data science.

In DS 110, students will use Python to explore many of the fundamental computer science concepts and processes used in data science. This begins with core computer science topics such as data structures, development of functions and recursion, and leads to topics including object oriented programming and data visualization.

In terms of Python tools for data science, students will learn how to work with numpy, pandas, and matplotlib to analyze real-world data. (See the more comprehensive list of topics below in this syllabus.)

The sequence of DS 110 and 210 works in concert with the 120-121-122 series: as students develop their expertise in the mathematical foundations covered in 120-121-122, they build their abilities to implement those tools and manage actual datasets in the 110-210 sequence.

Hub Learning Outcomes - Quantitative Reasoning I (QR1):

Students will be introduced to the Python programming language. Students will learn how to use pure Python to write basic functional and object-oriented programs. Students will also learn about core data science libraries such as numpy and pandas and, by the end of the course, students will be able to use these tools to produce descriptive analyses and visualizations of different datasets.

In parallel to learning Python for data science, students will develop and demonstrate their understanding of core concepts in computer science, and their ability to use them to manipulate different types of data. Students will learn about basic CS topics such as algorithms, big-O notation and algorithm runtime, object oriented programming methods, and parallel computing.

Students will master the fundamentals of programming in Python for data science, including learning how to use common libraries such as numpy, pandas, and matplotlib. Using these tools, students will build basic models of the data to uncover and interpret interesting patterns in real-world data.

Instructional Format, Course Pedagogy, and Approach to Learning

The course will be primarily lecture-based, but with additional exercises for the audience designed to engage and check understanding, including weekly for-credit quizzes. Homework will play a key role in moving from hypothetical concepts to applied problems. The discussion section will serve as additional review. The optional project will be a chance to let the students guide their own learning toward a personally meaningful goal.

Because of the prevalence of LLMs and other online resources that may mask gaps in a student's knowledge, understanding, and skills, two in-class midterms and a final will be administered in which other resources are not allowed.

Books and Other Course Materials

Recommended Text: Deitel & Deitel, *Intro to Python for Computer Science and Data Science*, ISBN-13 978-0135404676

Recommended readings by day are in the schedule at the end of this syllabus. This book can be helpful in providing an alternate take on the material, as well as in providing practice problems to help students prepare for the midterms. But, it isn't required.

Recommended Hardware: You need a basic laptop that has a web browser for this course. If you don't own one, you can [borrow one from BU](#). The heavy lifting of running programs will happen in the cloud, with very little done locally on your machine unless you prefer to work that way (see Anaconda below).

Courseware

We'll be using Blackboard for distributing assignments, collecting assignments, grading, and the distribution of course notes: <https://learn.bu.edu>

We will mostly be using Google Colab for assignments and course notes:
colab.research.google.com/

Students who wish to work offline could download and install Anaconda:
<https://www.anaconda.com/>

Students wanting help with the homework can try ChatGPT's new Study Mode:
<https://openai.com/index/chatgpt-study-mode/>

Students wanting additional practice programming problems before the exams can try our practice problem generator:
<https://huggingface.co/spaces/klgold/ds110probs>
(your 4-digit PIN is a column in Blackboard)

We'll use Piazza for Q&A and posting office hours:
<https://piazza.com/bu/fall2025/ds110/staff>

Assignments and Grading

Course assignments will consist mostly of programming, but will also contain some thought questions.

The individual assignments are meant to teach programming skills and other concepts. Students should try to do the work themselves when they can, because they will be tested on their abilities in the in-class exams, but AI, Piazza, or staff office hours can easily get students unstuck if they aren't making progress. Students are expected to honestly answer the question at the end of every homework about how much they relied on AI, but will not be penalized for using it.

In general, assignments are about all the topics that were covered in lecture since the last assignment - see the day-by-day topic list at the end of the syllabus for more details. All the assignments are weighted equally, even if maximum point values differ.

There will be one possible source of extra credit in the course - a mini-project that is part of assignment 8, where students can earn additional points by using a different dataset and/or performing additional analysis of the dataset. The best submissions will be invited to present a "lightning talk" near the end of the course for further extra credit.

There will be two in-class midterms and one final exam during the semester. One midterm will be at roughly the halfway point and will emphasize having learned to program, and the other exams will be cumulative and will include factual information about data science topics while still emphasizing programming. These exams will be partly multiple choice, partly free response coding. These exams are meant to encourage learning all the material rather than just what is emphasized on the homework, and also to discourage using generative AI in any way that impedes the learning process. Students should therefore practice on their own until they can produce basic programs quickly and without external aids.

Grading:

- In-class quizzes (10%)
- Midterm exam 1 (15%)
- Midterm exam 2 (20%)
- Individual assignments (15%, lowest dropped)
- Recitation participation (5%)
- Final exam (35%)

Resources/Support/How to Succeed in This Course:

1. To succeed in this course students should attend all lectures, come to discussion sections prepared with questions, and complete all assignments on time.
2. Students should discuss problems and material with fellow classmates or generative AI in a way that explores approaches instead of comparing answers.
3. Students are welcomed and encouraged to visit office hours to ask questions about the material covered in class, the homework, or any relevant topics.
4. The [Education Resource Center](#) offers free individual and group tutoring.
5. Accommodations for Students with Documented Disabilities: If you are a student with a disability or believe you might have a disability that requires accommodations, please contact the Office for Disability Services (ODS) at (617) 353-3658 or access@bu.edu to coordinate any reasonable accommodation requests. ODS is located at 25 Buick Street on the 3rd floor. Please give me notice of any special needs at least 2 weeks before an exam that requires special accommodations, so that we can schedule space if necessary.

Community of Learning: Class and University Policies

1. **Courtesy expectations.** Students are responsible for supporting a courteous learning environment. Please show respect for other students' questions, and maintain an attentive attitude in class. Please participate in the in-class exercises and activities. Using screens for activities other than following lecture is discouraged.
2. **Attendance & Absences.** Attending lecture is effectively mandatory because of the in-class quizzes that could be delivered on any non-exam day. Attending discussion sections also carries some participation credit. However, attendance or quizzes in either lecture or section can be waived with a reasonable request for accommodation. We will in particular waive for religious holidays; see the University [Policy on Religious Observance](#).
3. **Assignment Completion & Late Work.** Assignments will be submitted as PDF and .ipynb files on Blackboard. Students have 5 late days which can be used over the course of the semester, no more than 2 per assignment, which allow work to be turned in late with no penalty. Using 1 late day extends the deadline 24 hours for that student; late days can't be used in fractions.

4. Academic Conduct Statement

Students are expected to abide by the guidelines and rules of the Academic Code of Conduct. <https://www.bu.edu/academics/policies/academic-conduct-code/>

In particular, **if code is submitted that is heavily based on code found online or code produced by an AI, the source must be cited in the comments or a text box.** This is true for all submitted code (e.g., including the project). **Failure to cite a source may be considered plagiarism and reported to CDS for a hearing.**

In lieu of the CDS GAIA policy, we ask that students simply complete a one-question survey about how much AI they used, at the end of each assignment. We are no longer requiring transcripts of AI interactions.

You can consult the CS department's helpful webpage about code plagiarism if you have any doubts about what is considered ethical: <https://www.bu.edu/cs/undergraduate/undergraduate-life/academic-integrity/>

Outline of Class Meetings: Topics by Day, Optional Readings, Assignments Due

Normal homework goes out and is due on *Wednesdays* unless otherwise noted.

Week of 9/1 (BU classes start 9/2)

W Introduction

F Programming basics - expressions and types (Deitel 1.10, 2.3, 2.4)

Week of 9/8

M Variables and conditional branching (Deitel 2.2, 2.6, 2.7)

W while and lists (Deitel 3.7, 5.2, 5.8) *HW1 OUT*

F boolean operators, importing, and debugging (Deitel 3.14, 3.16)

Week of 9/15

M foreach loops; range (Deitel Ch 3.7-3.13)

W More on loops *HW1 DUE, HW2 OUT*

F Defining a function; typical commenting; refactoring; writing a test (Deitel Ch 4)

Week of 9/22

M More on functions

W Hash tables, sets, and dictionaries; pass by reference (Deitel Ch 6), *HW2 DUE, HW3 OUT*

F numpy

Week of 9/29

M matplotlib

W Writing bigger programs *HW3 DUE, HW4 OUT*

F Reading data into Pandas; escape characters; easy analysis in Pandas (Deitel Ch 7)

Week of 10/6

M More Pandas examples

W Strings, regexes, and data cleaning (Deitel Ch 8) *HW4 DUE, HW5 OUT*

F Files and Exceptions (Deitel Ch 9)

Week of 10/13

M Object-oriented programming: creating our own objects (Deitel Ch 10)

W Midterm review, *HW5 DUE*

F Midterm 1

Week of 10/20

M Common methods to override; other details of OO Python

W Recursion *HW6 OUT*

F Data structure implementation: trees, linked lists, dynamic arrays

Week of 10/27

M scikit-learn and k-nearest neighbors (Deitel Ch 15)

W Decision Trees *HW6 DUE, HW7 OUT*

F Decision Trees 2

Week of 11/3

M Decision Forests

W Regression *HW7 DUE, HW8 OUT*

F Using ML with Natural Language

Week of 11/10

M Complexity and big-O (Deitel Ch 11)

W Analysis of algorithms: sorts *HW8 DUE, HW9 OUT*

F Graphs and centrality

Week of 11/17

M SQL

W Midterm review, *HW9 DUE*

F Midterm 2

Week of 11/24

M Advanced Pandas

W,F Thanksgiving Break

Week of 12/1

M Visualization

W Other ways to speed up code *HW10 OUT*

F Effective communication

Week of 12/8

M Mini-Project Lightning Talks

W Final review day *HW10 DUE*

Dec 15-19 FINAL EXAM DAY TBD

~Last updated 8/19/25~